

Coding & Solution

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/TayyibahShaik/OptiCrop_Agriculture.git
Programming Language(s)	Python, JavaScript
Framework(s) Used	Flask (Backend), HTML/CSS/JS (Frontend)
Key Features Implemented	- Crop recommendation using ML model - User input form (NPK, temperature, humidity, pH, rainfall) - Real-time prediction output - Basic input validation
Pending / Incomplete Features	- Weather API integration - Fertilizer recommendation - Advanced UI improvements
Setup / Run Instructions	1. Install Python and required libraries (Flask, pandas, scikit-learn) 2. Run backend using python app.py 3. Open frontend in browser 4. Enter input values to get prediction

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The OptiCrop system successfully demonstrates the use of machine learning for agricultural decision-making.
- The application is functional and meets core requirements.
- Further improvements can be made in UI design and additional feature integration.